TIC 975879954

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

```
BLS EPHEMERIS
  Period       4.536910 d
  Epoch (t0)   1600.2829 BTJD
  Duration     0.1700 d
  Depth        2.043 %  (20,426 ppm)
  SNR          54.4

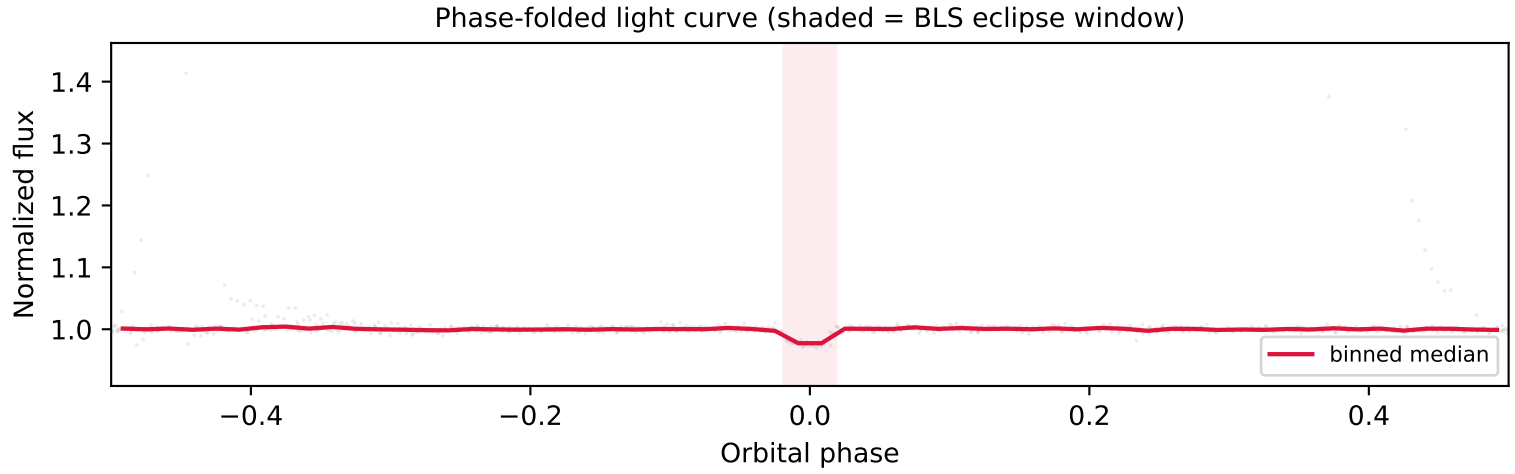
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     5  (gate >= 3)
  Robust depth ratio    1.10  (gate >= 0.50)
  Robust depth sigma    34.9  (gate >= 7.0)
  Odd/even z-score      0.15  (consistent)
  Secondary eclipse     not detected
```

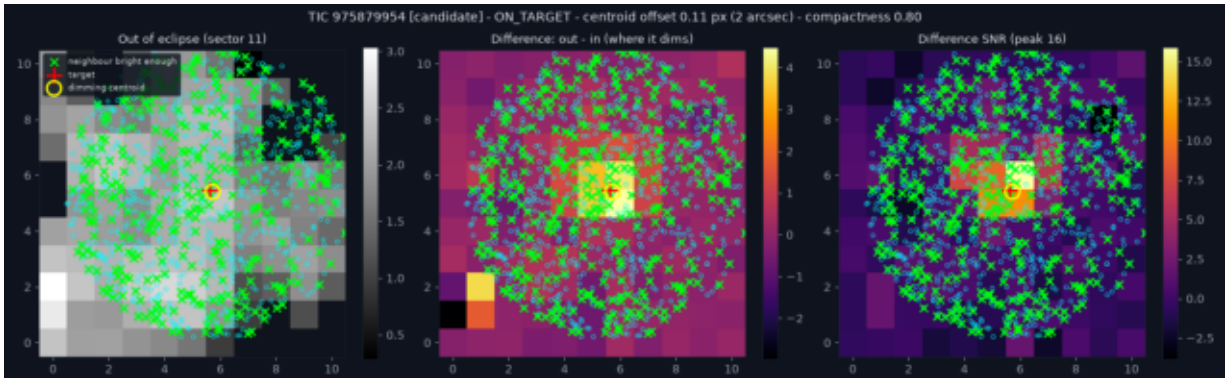
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 11    light-curve dip in this sector: 32.7 sigma    t0 refined by 0.0025 phase
  Difference-image peak SNR   15.92   (gate >= 5.0)      Compactness   0.80   (gate >= 0.50)
  Dimming centroid offset     0.11 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 309033686
    0.17 px away, dTmag +1.77 - farther from the dimming than the target
```





Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.062 px (dimming centroid 0.112 px from target vs 0.174 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.